

## Exercise 01

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## 1 Monotone Submodular Maximization

Consider a set  $\mathcal{U}$  of  $n$  elements that we can buy and a function  $f : 2^{\mathcal{U}} \rightarrow \mathbb{R}^+$ , where for each subset  $S \subseteq \mathcal{U}$ , the value  $f(S)$  determines our profit if we buy exactly the elements of set  $S$ .

We assume two properties about this profit function: (A) Function  $f$  is monotone in the sense that  $f(S) \leq f(T)$  for any two sets  $S, T$  such that  $S \subseteq T$ , and (B) Function  $f$  is submodular in the sense that  $f(S \cup \{i\}) - f(S) \geq f(T \cup \{i\}) - f(T)$  for any  $i \in \mathcal{U}$  and any two sets  $S, T$  such that  $S \subseteq T$ . In simple words, the submodularity means that the marginal gain that we have by adding  $i$  to our purchase set diminishes as we move from one purchase set  $S$  to a superset of it  $T$ . That is, roughly speaking, the more that we already have in the purchase set, the less extra gain by adding an element to it.

Devise an efficient algorithm that purchases a set  $S$  of (approximately) maximum profit, subject to the constraint that  $|S| \leq k$ , for some given value  $k \in \{1, 2, 3, \dots, n\}$ . What approximation factor do you get?

## 2 2-Approximation for Knapsack (Vazirani 8.2)

In the knapsack problem (discussed in the class), discard all elements that are larger than the budget  $B$ , and then sort the remaining elements by decreasing ratio of profit to size, let this order be  $a_1, a_2, \dots, a_n$ . Let  $k$  be the smallest number such that the total size of the first  $k$  elements  $a_1, a_2, \dots, a_k$  exceeds the budget  $B$ . Pick the more profitable of the following two options:  $\{a_1, a_2, \dots, a_{k-1}\}$  and  $\{a_k\}$ . Prove that this gives a 2-approximation for the most profitable set that fits in the knapsack.

## 3 Bin Covering (Vazirani 9.7)

Given  $n$  items with sizes  $a_1, a_2, \dots, a_n \in [c, 1]$  for some fixed constant  $c \in (0, 1)$ , give a Polynomial-Time Approximation Scheme (PTAS) for the problem of maximizing the number of bins, subject to the constraint that each bin has items with total size at least 1.